

Tagbo Ugo-Obi

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Summary: Master's student in Computer Science at Northwestern with prior experience in capital markets and M&A. Building expertise in computer vision, machine learning, and high-performance computing for engineering and research roles

EDUCATION

Northwestern University, McCormick School of Engineering

Evanston, IL

Master of Science in Computer Science

Expected Dec. 2027

- **Teaching Assistant** – Sports, Technology, and Learning
- **NU AI Safety Fellowship** (2026 – 2027)
- **Selected Coursework:** Machine Learning, Microcontroller Design, Parallel Computing, Parallel Architectures, Robotic Manipulation

University of Rochester

Rochester, NY

Bachelor of Arts in Statistics

- **Teaching Assistant** – Elements of Probability & Math Statistics; Computer Applications
- **Division III Varsity Football (LB)**

PROFESSIONAL EXPERIENCE

FHN Financial

Chicago, IL

Capital Markets – MBS Strategies Associate

Jan. 2021 – May 2025

- Conducted research and analysis on mortgage-backed securities, including agency and non-agency MBS, to support investment decisions and risk management activities
- Developed and back-tested trading strategies based on interest rate movements, housing market trends, and borrower behavior
- Produced market analysis, portfolio performance reporting, and investment recommendations for senior management and clients

Ziegler

Nashville, TN

Healthcare Corporate Finance – Investment Banking Analyst

Oct. 2018 – Jan. 2020

- Supported 5 sell-side M&A transactions through valuation analysis, due diligence, buyer research, financial analysis, and transaction execution
- Developed transaction materials and analyzed company operating data, using Excel to extract key metrics from raw billing data
- Coordinated information flow among clients, prospective buyers, and internal deal teams

PROJECTS

Parallel Sobel Edge Detection Pipeline (C / CUDA)

- Developed a multithreaded Sobel edge-detection pipeline using producer-consumer task scheduling and shared/partitioned ring-buffer work queues
- Evaluated scalability, throughput, and synchronization overhead across thread counts and shared versus partitioned queue configurations
- Implemented an exploratory CUDA version using persistent device buffers to compare CPU and GPU execution

Teleoperation Sensing & Communication (C)

- Integrated multiple IMUs with nRF52833 microcontrollers to capture wrist, forearm, and upper-arm orientation data over I2C and UART
- Implemented IEEE 802.15.4 sender/receiver communication to transmit sensor measurements for a team teleoperation system

Image Processing Algorithms (C)

- Implemented grayscale conversion, thresholding, image sharpening, and Sobel edge detection using pixel-level and neighborhood-based operations
- Applied convolution-based filtering and gradient calculations to transform images and extract edges and other visual features

Dynamic Memory Allocator (C)

- Implemented a simplified memory allocator in C with pool-based allocation, alignment, and block management
- Validated allocation and deallocation across variable block sizes to ensure correct memory management and prevent errors or leaks

Stock Exchange (Java)

- Developed an object-oriented stock exchange simulation in Java, implementing order management, trading books, and market-making functionality
- Applied Singleton, Façade, Flyweight, and Observer design patterns to structure trading-system components and interactions

TECHNICAL SKILLS

Languages: Java (Strong), C++ (Strong), C (Strong), VBA (Intermediate), Python (Familiar), R (Familiar)

Databases: MySQL (Intermediate)

Technologies: YieldBook, Bloomberg, S&P Capital IQ, Excel, PowerPoint